**END TERM EXAMINATION**

**(CBCS)(SUBJECTIVE TYPE)**

**Course Name: B.TECH(CSE AI/ AI ML) , Semester:2nd**

**(MAY, 2024)**

| Subject Code: BAI-102 | Subject: Object Oriented Programming using Java |
|---|---|
| Time :3 Hours | Maximum Marks :60 |

**Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.**

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| Q1 | | (5*4=20) | |
| (a) | What is the use of inheritance and explain the parameterized methods in a class. | | |
| (b) | Write a program to swap two numbers( say n1 and n2) and display the numbers before and after swapping. | | |
| (c) | Explain the use of variables and methods of an interface and an abstract class. | | |
| (d) | Differentiate between AWT Components and Containers. Explain the use of sleep in Threading. | | |
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| | **UNIT-I** | | |
| Q2 | Explain the constructor and its use.Create a class Complex(member variables real , imaginary) ,which includes appropriate constructor and methods to initialize the members. Provide a method to Perform addition of any two complex numbers and display the result and create three objects using an array of objects. | (10) | |
| Q3 | What are local variables, instance variables and class variables in java?How to create an array in java?Write a program in java to find the minimum, maximum and average of elements in an array. | (10) | |
| | **UNIT-II** | | |
| Q4 | Create a program to explain method overloading, method overriding and constructor overloading. Create a class MyDate( member variables:day, month and year) for manipulating dates. Provide a constructor that enables an object of this class to be initialized when it is declared (You can select any default values for the day, month & year). Provide a main function which will create 3 objects of Date class using an array of objects. | (10) | |
| Q5 | Explain package and its use in java.Define a class Tour in a package pack1(Tour.java) with Members tourcode(type:integer),noofkids(type:integer) ,tourfare(type:integer).Provide appropriate constructor and methods to initialize and display the details of a tour.Create 3 objects in the different class(TourMain.java) in another package pack2. | (10) | |
| | **UNIT-III** | | |
| Q6 | Explain the role of try,catch,throw,throws and finally for handling exceptions.Using programs explain any two types of exceptions and how to handle them. | (10) | |
| Q7 | What is I/O stream and Object Serialization? Design a program to read and write from a file and handle exceptions. | (10) | |
| | **UNIT-IV** | | |
| Q8 | Explain the life cycle of thread?Why do we use Runnable Interface?Create a program in java to show the use of multiple threads. | (10) | |
| Q9 | What is the role of event, event source and event listeners in the event handling model?Create a container,change the layout of the container and add at least five different components and perform event handling on any two components. | (10) | |